Multiple regression analysis with qualitative information

Econometrics. Lecture 5

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3. Slope dummy variables
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1. The nature of qualitative information

Describing qualitative information

Qualitative factors often come in the form of binary information. A **binary variable** (or **dummy variable**) takes only two values – zero and one.

Examples:

- gender may play a very important role in determining salaries;
- different ethnic groups may follow different behavioral patterns;
- educational levels can affect earnings from employment;
- changes in a political regime may affect economic conditions;
- a war can have an impact on all aspects of economic activity;
- a country income level may be associated with different reaction to shocks.

2. Intercept dummy variables

A single dummy independent variable

A simple model of hourly wage determination

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u$$

where

$$female = \begin{cases} 1 & \text{when the person is female} \\ 0 & \text{when the person is male} \end{cases}$$

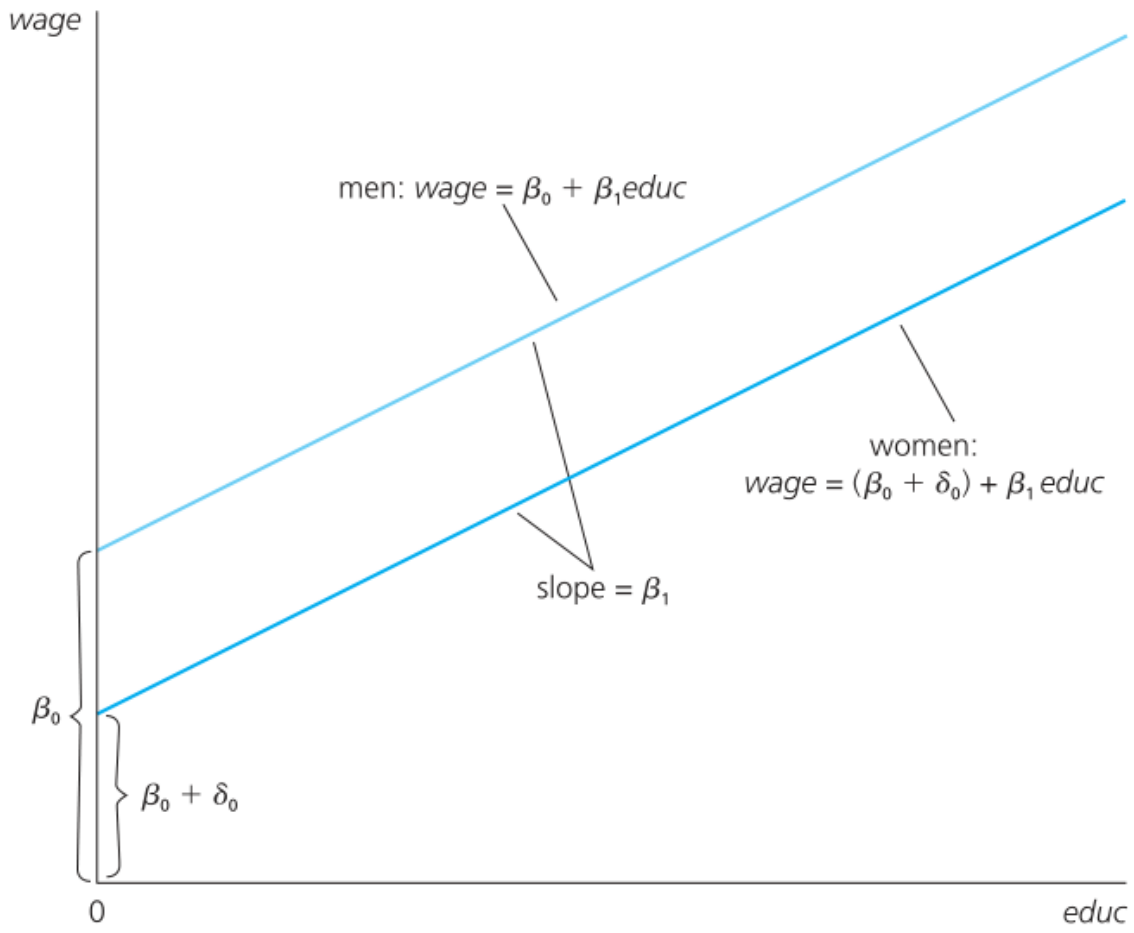
Males is the **base group** or **benchmark group**, that is, the group against which comparisons are made.

β_0 is the intercept for males.

δ_0 is the difference in wage between females and males, given the same amount of education (and the same error term *u*). The intercept for females is $\beta_0 + \delta_0$.

An intercept shift

An intercept shift between males and females



$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u$$

$$female = \begin{cases} 1 & \text{when the person is female} \\ 0 & \text{when the person is male} \end{cases}$$

$$male = \begin{cases} 1 & \text{when the person is male} \\ 0 & \text{when the person is female} \end{cases}$$

Question: How many dummy variables can we include in a model with an intercept?

Answer: Only one! Using two dummy variables would introduce perfect collinearity:

$$female + male = 1$$

Wage discrimination

An extended model of hourly wage determination

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

The null hypothesis of no difference between men and women is

$$H_0: \delta_0 = 0$$

The alternative that there is discrimination against women is

$$H_1: \delta_0 < 0$$

p-value = 0.00 < 0.05

We reject $H_0: \delta_0 = 0$.

Dependent Variable: WAGE
Method: Least Squares
Date: 01/04/26 Time: 16:38
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-1.567939	0.724551	-2.164014	0.0309
FEMALE	-1.810852	0.264825	-6.837915	0.0000
EDUC	0.571505	0.049337	11.58362	0.0000
EXPER	0.025396	0.011569	2.195083	0.0286
TENURE	0.141005	0.021162	6.663225	0.0000

R-squared	0.363541	Mean dependent var	5.896103
Adjusted R-squared	0.358655	S.D. dependent var	3.693086
S.E. of regression	2.957572	Akaike info criterion	5.016075
Sum squared resid	4557.308	Schwarz criterion	5.056619
Log likelihood	-1314.228	Hannan-Quinn criter.	5.031950
F-statistic	74.39801	Durbin-Watson stat	1.794400
Prob(F-statistic)	0.000000		

If we take a woman and a man with the same levels of education, experience, and tenure, the woman earns, on average, $1.81 less per hour than the man.

Dummy explanatory variables when the dependent variable is log(*y*)

Question: How do we interpret the dummy variable coefficient when the dependent variable appearing in logarithmic form?

When log(*y*) is the dependent variable in a model, the coefficient on a dummy variable, when multiplied by 100, is interpreted as the *percentage difference* in *y*, holding all other factors fixed.

Generally, if $\hat{\beta}_1$ is the coefficient on a dummy variable, say x_1, when log(*y*) is the dependent variable, the exact percentage difference in the predicted *y* when $x_1 = 1$ versus when $x_1 = 0$ is

$$100 * [\exp(\hat{\beta}_1) - 1]$$

Wage discrimination when the dependent variable is log(*wage*)

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/04/26 Time: 16:56
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.501348	0.101902	4.919885	0.0000
FEMALE	-0.301146	0.037246	-8.085414	0.0000
EDUC	0.087462	0.006939	12.60463	0.0000
EXPER	0.004629	0.001627	2.845087	0.0046
TENURE	0.017367	0.002976	5.835235	0.0000

R-squared	0.392270	Mean dependent var	1.623268
Adjusted R-squared	0.387604	S.D. dependent var	0.531538
S.E. of regression	0.415959	Akaike info criterion	1.093001
Sum squared resid	90.14445	Schwarz criterion	1.133545
Log likelihood	-282.4592	Hannan-Quinn criter.	1.108876
F-statistic	84.07213	Durbin-Watson stat	1.775217
Prob(F-statistic)	0.000000		

$$\hat{\delta}_0 = -0.301$$

For the same levels of education, experience, and tenure, women earn approximately 30.1% less than men.

The more accurate estimate implies that a woman's wage is, on average, 26% below a comparable man's wage.

$$\exp(\hat{\delta}_0) - 1 = \exp(-0.301) - 1 \approx -0.260$$

Interactions (1)

$$female = \begin{cases} 1 & \text{when the person is female} \\ 0 & \text{when the person is male} \end{cases}$$

$$married = \begin{cases} 1 & \text{when the person is married} \\ 0 & \text{when the person is single} \end{cases}$$

An **interaction term**

$$female * married$$

$$\log(wage) = \beta_0 + \beta_1 female + \beta_2 married + \beta_3 female * married + \\ + \beta_4 educ + \beta_5 exper + \beta_6 tenure + u$$

Interactions (2)

$$\log(wage) = \beta_0 + \beta_1 female + \beta_2 married + \beta_3 female * married + \\ + \beta_4 educ + \beta_5 exper + \beta_5 tenure + u$$

$\beta_0 = 0.388$ – an intercept for a single men group (*female* = 0 and *married* = 0)

$\beta_1 \approx -9.7\%$ – a proportional difference in wage between a single woman and a single man (*female* = 1 and *married* = 0).

$\beta_2 \approx 29.2\%$ – a proportional difference in wage between a married man and a single man (*female* = 0 and *married* = 1).

$\beta_1 + \beta_2 + \beta_3 \approx -12\%$ – a proportional difference in wage between a married woman and a single man (*female* = 1 and *married* = 1)

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/04/26 Time: 19:27
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.387817	0.102322	3.790171	0.0002
FEMALE	-0.096750	0.057463	-1.683681	0.0928
MARRIED	0.292091	0.055389	5.273454	0.0000
FEMALE*MARRIED	-0.315576	0.073967	-4.266437	0.0000
EDUC	0.083533	0.006861	12.17532	0.0000
EXPER	0.003164	0.001655	1.911744	0.0565
TENURE	0.015684	0.002921	5.369541	0.0000

R-squared	0.423819	Mean dependent var	1.623268
Adjusted R-squared	0.417158	S.D. dependent var	0.531538
S.E. of regression	0.405798	Akaike info criterion	1.047296
Sum squared resid	85.46477	Schwarz criterion	1.104059
Log likelihood	-268.4389	Hannan-Quinn criter.	1.069521
F-statistic	63.62646	Durbin-Watson stat	1.774109
Prob(F-statistic)	0.000000		

Dummies for multiple categories

Estimated model with interaction term

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/04/26 Time: 19:27
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.387817	0.102322	3.790171	0.0002
FEMALE	-0.096750	0.057463	-1.683681	0.0928
MARRIED	0.292091	0.055389	5.273454	0.0000
FEMALE*MARRIED	-0.315576	0.073967	-4.266437	0.0000
EDUC	0.083533	0.006861	12.17532	0.0000
EXPER	0.003164	0.001655	1.911744	0.0565
TENURE	0.015684	0.002921	5.369541	0.0000

R-squared	0.423819	Mean dependent var	1.623268
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Sum squared resid	85.46477	Schwarz criterion	1.104059
Log likelihood	-268.4389	Hannan-Quinn criter.	1.069521
F-statistic	63.62646	Durbin-Watson stat	1.774109
Prob(F-statistic)	0.000000		

Estimated model with dummies for groups

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/04/26 Time: 20:30
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.387817	0.102322	3.790171	0.0002
SINGFEM	-0.096750	0.057463	-1.683681	0.0928
MARMALE	0.292091	0.055389	5.273454	0.0000
MARFEM	-0.120235	0.057984	-2.073577	0.0386
EDUC	0.083533	0.006861	12.17532	0.0000
EXPER	0.003164	0.001655	1.911744	0.0565
TENURE	0.015684	0.002921	5.369541	0.0000

R-squared	0.423819	Mean dependent var	1.623268
Adjusted R-squared	0.417158	S.D. dependent var	0.531538
S.E. of regression	0.405798	Akaike info criterion	1.047296
Sum squared resid	85.46477	Schwarz criterion	1.104059
Log likelihood	-268.4389	Hannan-Quinn criter.	1.069521
F-statistic	63.62646	Durbin-Watson stat	1.774109
Prob(F-statistic)	0.000000		

Four groups: 1) single men (a base group); 2) married men (*marmale*), 3) married women (*marfem*), 4) single women (*singfem*).

3. Slope dummy variables

Allowing for different slopes

We wish to test whether the return to education is the same for men and women, allowing for a constant wage differential between men and women.

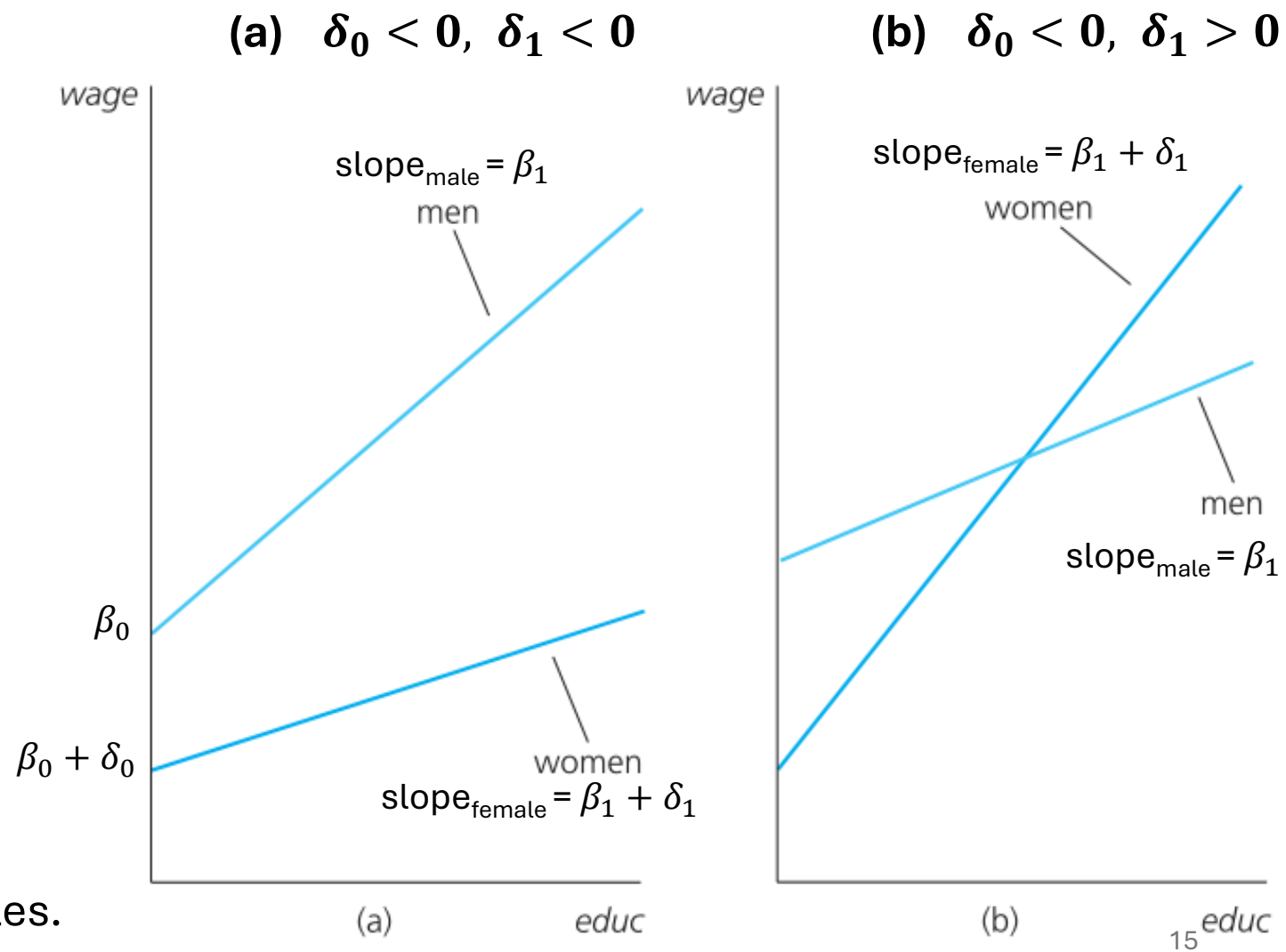
$$\log(wage) = \beta_0 + \delta_0 female + \\ + \beta_1 educ + \delta_1 female * educ + u$$

β_0 – the intercept for males.

$\beta_0 + \delta_0$ – the intercept for females.

β_1 – the slope on education for males.

$\beta_1 + \delta_1$ – the slope on education for females.



Wage equation with different slopes

$$\log(wage) = \beta_0 + \delta_0 female + \beta_1 educ + \\ + \delta_1 female * educ + \beta_2 exper + \beta_3 tenure + u$$

$H_0: \delta_1 = 0$ vs. $H_1: \delta_1 < 0$

(the return to education is the same for women and men)

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 10:49
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.464712	0.122892	3.781465	0.0002
FEMALE	-0.210376	0.173969	-1.209273	0.2271
EDUC	0.090276	0.008715	10.35851	0.0000
FEMALE*EDUC	-0.007245	0.013562	-0.534164	0.5935
EXPER	0.004642	0.001628	2.850310	0.0045
TENURE	0.017436	0.002981	5.848873	0.0000

R-squared	0.392603	Mean dependent var	1.623268
Adjusted R-squared	0.386763	S.D. dependent var	0.531538
S.E. of regression	0.416245	Akaike info criterion	1.096254
Sum squared resid	90.09501	Schwarz criterion	1.144908
Log likelihood	-282.3149	Hannan-Quinn criter.	1.115304
F-statistic	67.22251	Durbin-Watson stat	1.774145
Prob(F-statistic)	0.000000		

p-value = 0.59 > 0.05

We fail to reject $H_0: \delta_1 = 0$.

We are also interested in the hypothesis that average wages are identical for men and women who have the same levels of education.

$H_0: \delta_0 = 0, \delta_1 = 0$ vs. $H_1: \delta_0 \neq 0 \; or \; \delta_1 \neq 0$

Redundant Variables Test
Equation: EQ_504
Redundant variables: FEMALE FEMALE*EDUC
Specification: LOG(WAGE) C FEMALE EDUC FEMALE*EDUC EXPER TENURE
Null hypothesis: FEMALE FEMALE*EDUC are jointly insignificant

	Value	df	Probability
F-statistic	32.78479	(2, 520)	0.0000
Likelihood ratio	62.46576	2	0.0000

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	11.36056	2	5.680281
Restricted SSR	101.4556	522	0.194359
Unrestricted SSR	90.09501	520	0.173260

p-value = 0.00 < 0.05

We reject H_0

Wage equation with different slopes. Transforming the variables

The differential at zero years of education (model 1) is not informative. More interesting would be to estimate the gender differential at the average education level in the sample (model 2).

Estimated model 1

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 10:49
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.464712	0.122892	3.781465	0.0002
FEMALE	-0.210376	0.173969	-1.209273	0.2271
EDUC	0.090276	0.008715	10.35851	0.0000
FEMALE*EDUC	-0.007245	0.013562	-0.534164	0.5935
EXPER	0.004642	0.001628	2.850310	0.0045
TENURE	0.017436	0.002981	5.848873	0.0000

R-squared	0.392603	Mean dependent var	1.623268
Adjusted R-squared	0.386763	S.D. dependent var	0.531538
S.E. of regression	0.416245	Akaike info criterion	1.096254
Sum squared resid	90.09501	Schwarz criterion	1.144908
Log likelihood	-282.3149	Hannan-Quinn criter.	1.115304
F-statistic	67.22251	Durbin-Watson stat	1.774145
Prob(F-statistic)	0.000000		

Estimated model 2

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 11:53
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.464712	0.122892	3.781465	0.0002
FEMALE	-0.301388	0.037274	-8.085756	0.0000
EDUC	0.090276	0.008715	10.35851	0.0000
FEMALE*(EDUC-@MEAN(EDUC,"1 526"))	-0.007245	0.013562	-0.534164	0.5935
EXPER	0.004642	0.001628	2.850310	0.0045
TENURE	0.017436	0.002981	5.848873	0.0000

R-squared	0.392603	Mean dependent var	1.623268
Adjusted R-squared	0.386763	S.D. dependent var	0.531538
S.E. of regression	0.416245	Akaike info criterion	1.096254
Sum squared resid	90.09501	Schwarz criterion	1.144908
Log likelihood	-282.3149	Hannan-Quinn criter.	1.115304
F-statistic	67.22251	Durbin-Watson stat	1.774145
Prob(F-statistic)	0.000000		

Effects of race on baseball player salaries

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/05/26 Time: 12:31
Sample: 1 353
Included observations: 330

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.55329	0.130457	88.56011	0.0000
YEARS	0.066751	0.012707	5.253125	0.0000
GAMESYR	0.011073	0.002214	5.001141	0.0000
RBISYR	0.011830	0.003504	3.375798	0.0008
ALLSTAR	0.008788	0.002716	3.236138	0.0013
BLACK	-0.160710	0.122572	-1.311149	0.1907
HISPAN	-0.200180	0.152703	-1.310911	0.1908
BLACK*PERCBLCK	0.012754	0.004930	2.586868	0.0101
HISPAN*PERCHISP	0.020711	0.009779	2.117836	0.0350

R-squared	0.633437	Mean dependent var	13.51172
Adjusted R-squared	0.624301	S.D. dependent var	1.163087
S.E. of regression	0.712905	Akaike info criterion	2.187958
Sum squared resid	163.1431	Schwarz criterion	2.291569
Log likelihood	-352.0130	Hannan-Quinn criter.	2.229287
F-statistic	69.33771	Durbin-Watson stat	1.317453
Prob(F-statistic)	0.000000		

Redundant Variables Test
Equation: EQ05
Redundant variables: BLACK HISPAN BLACK*PERCBLCK HISPAN *PERCHISP
Specification: LOG(SALARY) C YEARS GAMESYR RBISYR ALLSTAR BLACK HISPAN BLACK*PERCBLCK HISPAN*PERCHISP
Null hypothesis: BLACK HISPAN BLACK*PERCBLCK HISPAN*PERCHISP are jointly insignificant

	Value	df	Probability
F-statistic	2.887124	(4, 321)	0.0226
Likelihood ratio	11.66371	4	0.0200

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	5.869339	4	1.467335
Restricted SSR	169.0125	325	0.520038
Unrestricted SSR	163.1431	321	0.508234

The variables *black* and *hispan* are binary indicators for the individual players.

The variable *percblck* is the percentage of the team's city that is black, and *perchisp* is the percentage of Hispanics.

The other variables measure aspects of player productivity and longevity.

Summary

In the simplest case, a dummy variable is defined to distinguish between two groups, and the coefficient estimate on the dummy variable estimates the ceteris paribus difference between the two groups.

Allowing for more than two groups is accomplished by defining a set of dummy variables: if there are g groups, then $g–1$ dummy variables are included in the model.

All estimates on the dummy variables are interpreted relative to the base group (the group for which no dummy variable is included in the model).

Dummy variables can be interacted with quantitative variables to allow slope differences across different groups.

4. Binary dependent variable: Linear probability model

The linear probability model (LPM)

Sometimes the dependent variable is also binary (for example, whether a person is employed or unemployed).

Consider a MLRM where y is a binary variable:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$$

The multiple linear regression model with a binary dependent variable is called the **linear probability model (LPM)** because the response probability is linear in the parameters β_j:

$$\mathrm{P}(y = 1|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

The slope coefficient β_j measures the predicted change in the probability of success when x_j increases by one unit:

$$\Delta\mathrm{P}(y = 1|\mathbf{x}) = \beta_j \Delta x_j$$

Probability of “being in labor force”

Let *inlf* (“in the labor force”) be a binary variable indicating labor force participation by a married woman during 1975: *inlf* = 1 if the woman reports working for a wage outside the home at some point during the year, and zero otherwise.

Labor force participation depends on other sources of income, including husband’s earnings (*nwifeinc*, $ thousands), years of education (*educ*), past years of labor market experience (*exper*), age, number of children less than six years old (*kidslt6*), and number of kids between 6 and 18 years of age (*kidsge6*).

For example, the coefficient on *educ* means that, everything else is held fixed, another year of education increases the probability of labor force participation by 0.038.

10 more years of education increases the probability of being in the labor force by 0.038*10 = 0.38.

Dependent Variable: INLF
Method: Least Squares
Date: 07/27/26 Time: 11:51
Sample: 1 753
Included observations: 753

Variable	Coefficient	Std. Error	t-Statistic	Prob.
NWIFEINC	-0.003405	0.001448	-2.350841	0.0190
EDUC	0.037995	0.007376	5.151194	0.0000
EXPER	0.039492	0.005673	6.961866	0.0000
EXPER^2	-0.000596	0.000185	-3.226959	0.0013
AGE	-0.016091	0.002485	-6.476014	0.0000
KIDSLT6	-0.261810	0.033506	-7.813888	0.0000
KIDSGE6	0.013012	0.013196	0.986077	0.3244
C	0.585519	0.154178	3.797683	0.0002

R-squared	0.264216	Mean dependent var	0.568393
Adjusted R-squared	0.257303	S.D. dependent var	0.495630
S.E. of regression	0.427133	Akaike info criterion	1.147124
Sum squared resid	135.9197	Schwarz criterion	1.196251
Log likelihood	-423.8923	Hannan-Quinn criter.	1.166051
F-statistic	38.21795	Durbin-Watson stat	0.493840
Prob(F-statistic)	0.000000		

Problem 1: Heteroskedasticity

When *y* is a binary variable, its variance, conditional on *x*, is

$$\mathrm{Var}(y|\mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})]$$

This means that, except in the case where the probability does not depend on any of the independent variables, there must be heteroskedasticity in a linear probability model.

This does not cause bias in the OLS estimators of the β_j, but the estimators of their variances are biased. The solution is to use robust standard errors (covered in the topic 6).

Problem 2:
Probability can be negative or > 1

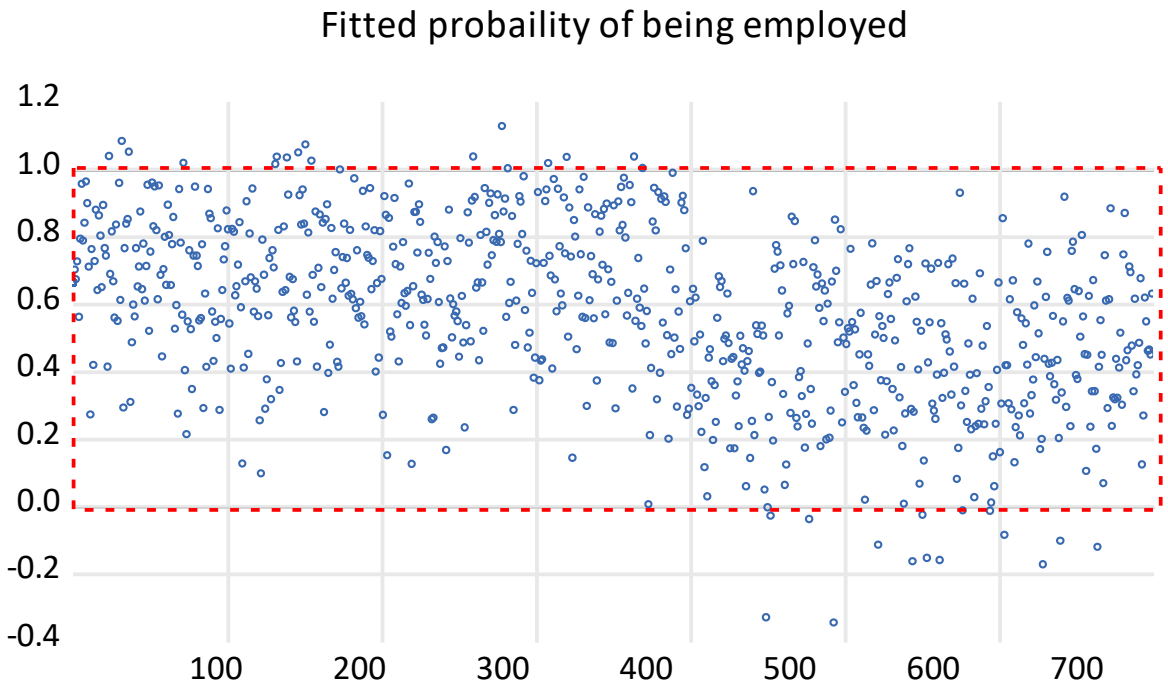
Consider the estimated model

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Date: 07/27/26 Time: 11:51
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Included observations: 753

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Prob(F-statistic)	0.000000		

Fitted (predicted) probabilities of being employed



Predicted probabilities outside the unit interval are a little troubling when we want to make predictions

Define a predicted value as

$\tilde{y}_i = 1$ if $\hat{y}_i \geq 0.5$ and $\tilde{y}_i = 0$ if $\hat{y}_i < 0.5$

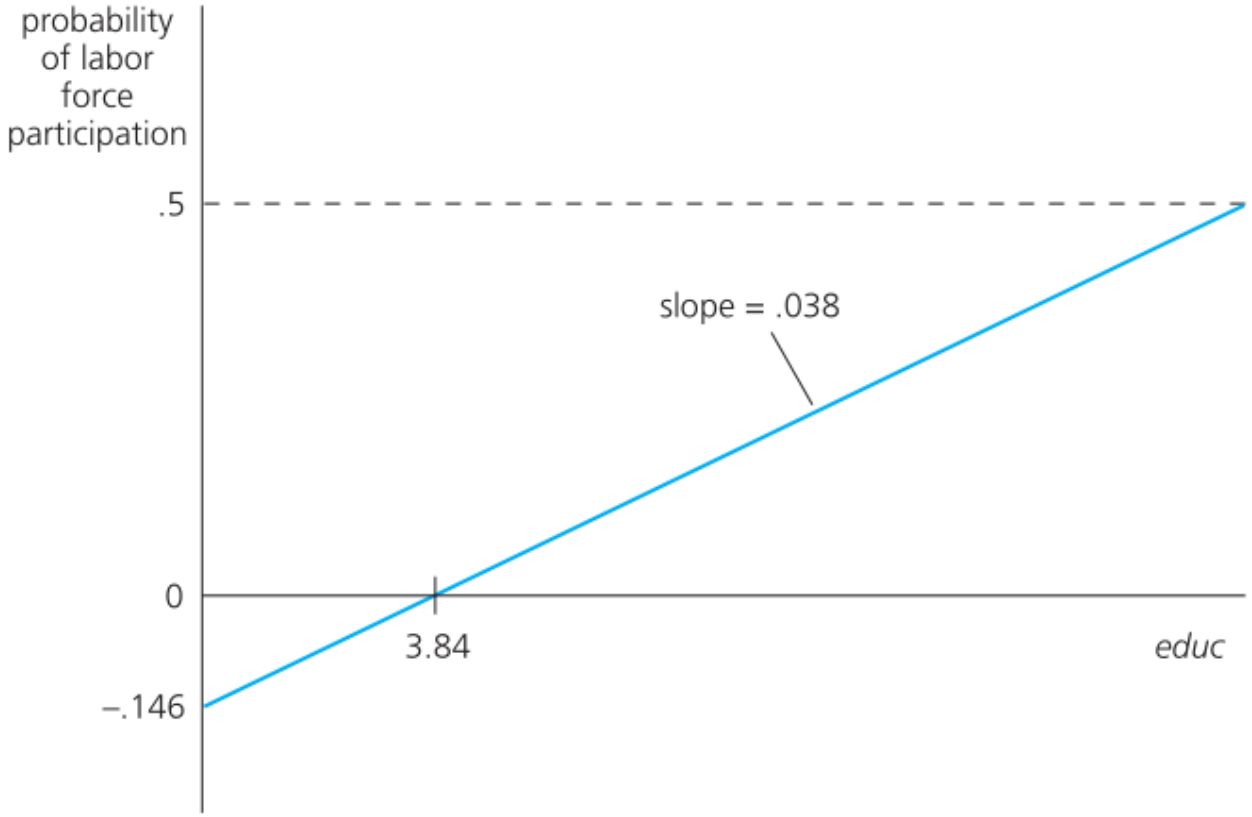
Use the data on y_i and $\tilde{y}_i$ to obtain the frequencies with which we correctly predict $y_i = 1$ and $y_i = 0$.

The proportion of overall correct predictions is a widely used goodness-of-fit measure for binary dependent variables: the **percent correctly predicted**.

In our case the percent correctly predicted is 73.4%.

Problem 3:
Partial effect of any explanatory variable (appearing in level form) is constant

Estimated relationship between the probability of being in the labor force and years of education, with other explanatory variables fixed



Solution: Use probit or logit

Summary

The linear probability model, which is simply estimated by OLS, allows us to explain a binary response using regression analysis.

The OLS estimates are now interpreted as changes in the probability of “success” ($y = 1$), given a one-unit increase in the corresponding explanatory variable.

The LPM does have some drawbacks: it can produce predicted probabilities that are less than zero or greater than one, it implies a constant marginal effect of each explanatory variable that appears in its original form, and it contains heteroskedasticity.

5. Binary dependent variable: Probit / Logit model

Nonlinear binary response models

Consider a class of binary response models of the form

$$\mathrm{P}(y = 1|\mathbf{x}) = G(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)$$

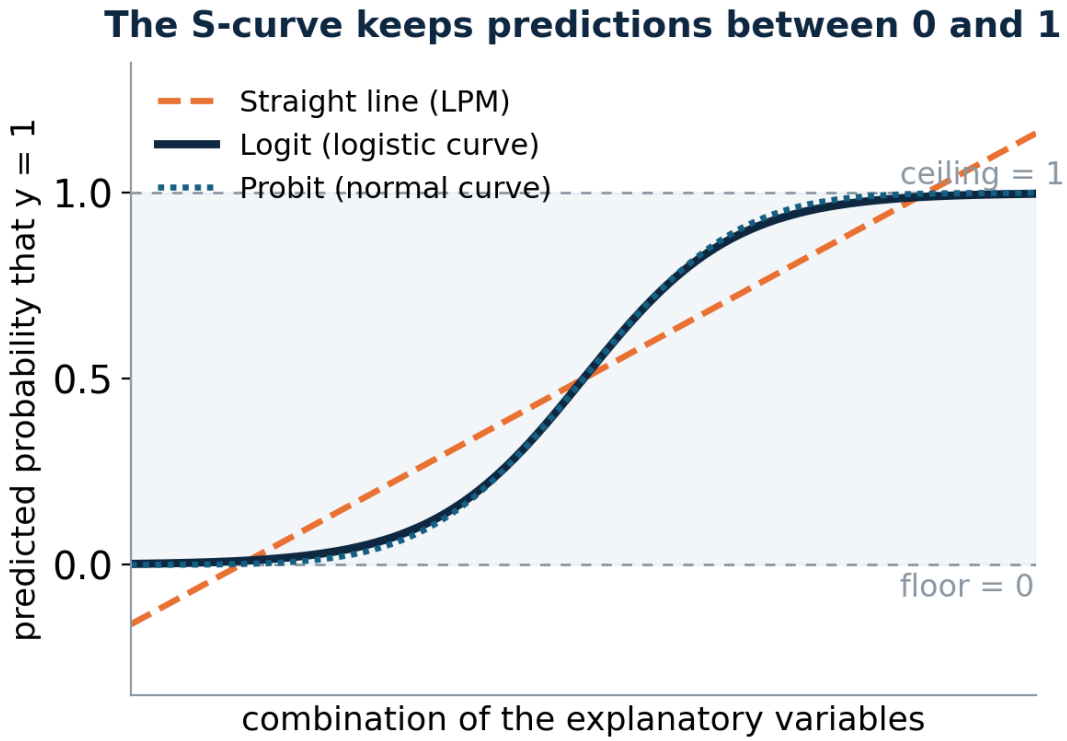
where G is a function taking on values strictly between zero and one.

In the **logit model**, G is the logistic function:

$$G(z) = \exp(z)/[1 + \exp(z)]$$

In the **probit model**, G is the standard normal cumulative distribution function:

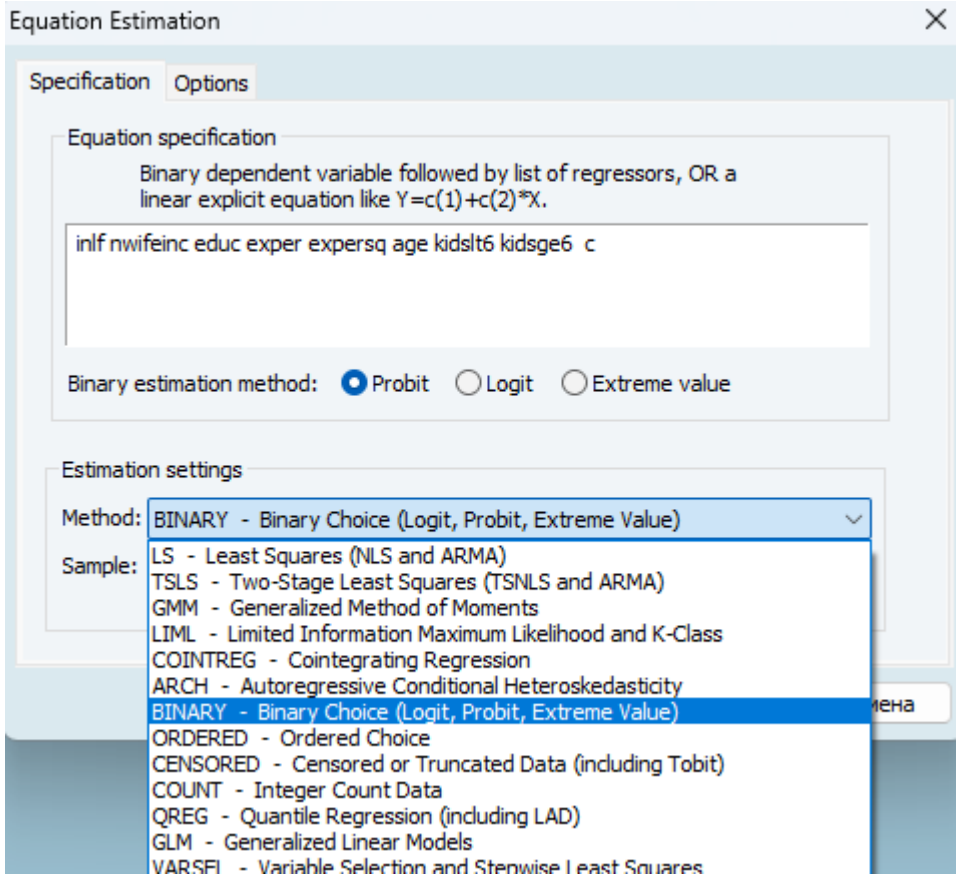
$$G(z) = \Phi\,(z) = \int_{-\infty}^{z} \phi(v)dv$$



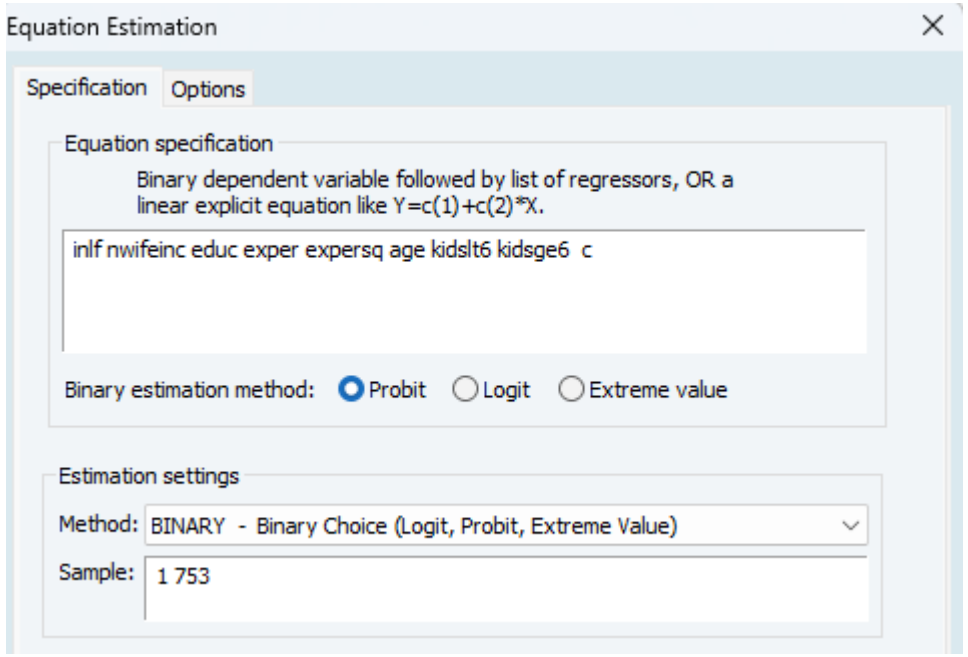
Maximum likelihood estimation (MLE) is used to build probit / logit models.

Logit and probit in EViews

Select the “BINARY” method



Choose “Probit” or “Logit”



Logit and probit in EViews. Estimation results

Probit

Dependent Variable: INLF
Method: ML - Binary Probit (Newton-Raphson / Marquardt steps)
Date: 07/27/26 Time: 13:09
Sample: 1 753
Included observations: 753
Convergence achieved after 4 iterations
Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
NWIFEINC	-0.012024	0.004840	-2.484327	0.0130
EDUC	0.130905	0.025254	5.183485	0.0000
EXPER	0.123348	0.018716	6.590348	0.0000
EXPERSQ	-0.001887	0.000600	-3.145205	0.0017
AGE	-0.052853	0.008477	-6.234656	0.0000
KIDSLT6	-0.868329	0.118522	-7.326287	0.0000
KIDSGE6	0.036005	0.043477	0.828142	0.4076
C	0.270077	0.508593	0.531027	0.5954

McFadden R-squared	0.220581	Mean dependent var	0.568393
S.D. dependent var	0.495630	S.E. of regression	0.425945
Akaike info criterion	1.087124	Sum squared resid	135.1646
Schwarz criterion	1.136251	Log likelihood	-401.3022
Hannan-Quinn criter.	1.106050	Deviance	802.6044
Restr. deviance	1029.746	Restr. log likelihood	-514.8732
LR statistic	227.1420	Avg. log likelihood	-0.532938
Prob(LR statistic)	0.000000		

Obs with Dep=0	325	Total obs	753
Obs with Dep=1	428		

Logit

Dependent Variable: INLF
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)
Date: 07/25/26 Time: 23:56
Sample: 1 753
Included observations: 753
Convergence achieved after 3 iterations
Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
NWIFEINC	-0.021345	0.008421	-2.534620	0.0113
EDUC	0.221170	0.043440	5.091442	0.0000
EXPER	0.205870	0.032057	6.422001	0.0000
EXPERSQ	-0.003154	0.001016	-3.104093	0.0019
AGE	-0.088024	0.014573	-6.040232	0.0000
KIDSLT6	-1.443354	0.203585	-7.089692	0.0000
KIDSGE6	0.060112	0.074790	0.803749	0.4215
C	0.425452	0.860370	0.494499	0.6210

McFadden R-squared	0.219681	Mean dependent var	0.568393
S.D. dependent var	0.495630	S.E. of regression	0.425963
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Restr. deviance	1029.746	Restr. log likelihood	-514.8732
LR statistic	226.2161	Avg. log likelihood	-0.533553
Prob(LR statistic)	0.000000		

Obs with Dep=0	325	Total obs	753
Obs with Dep=1	428		

Unlike the LPM, these coefficients are **not** the change in probability – their size cannot be read directly.

What you can read is the **sign**: a positive coefficient always means “more likely to work,” a negative one “less likely.”

The **relative effects** of any two continuous explanatory variables do **not depend** on ***x***: the ratio of the partial effects for x_j and x_h is β_j/β_h.

The magnitudes of the coefficient estimates across models are not directly comparable

LPM

Dependent Variable: INLF
Method: Least Squares
Date: 07/27/26 Time: 11:51
Sample: 1 753
Included observations: 753

Variable	Coefficient	Std. Error	t-Statistic	Prob.
NWIFEINC	-0.003405	0.001448	-2.350841	0.0190
EDUC	0.037995	0.007376	5.151194	0.0000
EXPER	0.039492	0.005673	6.961866	0.0000
EXPER^2	-0.000596	0.000185	-3.226959	0.0013
AGE	-0.016091	0.002485	-6.476014	0.0000
KIDSLT6	-0.261810	0.033506	-7.813888	0.0000
KIDSGE6	0.013012	0.013196	0.986077	0.3244
C	0.585519	0.154178	3.797683	0.0002

R-squared	0.264216	Mean dependent var	0.568393
Adjusted R-squared	0.257303	S.D. dependent var	0.495630
S.E. of regression	0.427133	Akaike info criterion	1.147124
Sum squared resid	135.9197	Schwarz criterion	1.196251
Log likelihood	-423.8923	Hannan-Quinn criter.	1.166051
F-statistic	38.21795	Durbin-Watson stat	0.493840
Prob(F-statistic)	0.000000		

Probit

Dependent Variable: INLF
Method: ML - Binary Probit (Newton-Raphson / Marquardt steps)
Date: 07/27/26 Time: 13:09
Sample: 1 753
Included observations: 753
Convergence achieved after 4 iterations
Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
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LR statistic	227.1420	Avg. log likelihood	-0.532938
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Obs with Dep=0	325	Total obs	753
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Logit

Dependent Variable: INLF
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)
Date: 07/25/26 Time: 23:56
Sample: 1 753
Included observations: 753
Convergence achieved after 3 iterations
Coefficient covariance computed using observed Hessian

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LR statistic	226.2161	Avg. log likelihood	-0.533553
Prob(LR statistic)	0.000000		

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Obs with Dep=1	428		

There are rough conversion rules:

- a logit coefficient is about **4x** the LPM one,
- a probit coefficient about **2.5x**,
- a logit coefficient is about **1.6x** the probit (the tightest rule).

The signs, the significance, and the relative sizes are identical across the three.

Interpretation of the estimates in logit and probit. Partial effect at the mean

- Calculate the partial effect at the mean.
  - Insert the mean values of all the explanatory variables into the estimated logit equation and calculate fitted probability at the mean values of the regressors.
  - Recalculate, but now increasing the value of the explanatory variable under examination by one unit to obtain the new fitted probability.
  - The difference between the two shows the partial (or marginal) effect at the mean.

This approach should be used cautiously when one or more of the explanatory variables is also a dummy variable. When dummies exist in the equation then set the dummy explanatory variable first equal to one and then equal to zero. And compare the results.

Interpretation of the estimates in logit and probit. Partial effect at the mean in EViews

Logit

Dependent Variable: INLF
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)
Date: 07/25/26 Time: 23:56
Sample: 1 753
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Prob(LR statistic)	0.000000		

Obs with Dep=0	325	Total obs	753
Obs with Dep=1	428		

- Calculate the partial effect of education at the mean.
  - Probability at the mean values of regressors:

```
scalar ave_p_mean = @logit(eq_logit.c(1)*@mean(NWIFEINC) +
eq_logit.c(2)*@mean(EDUC) + eq_logit.c(3)*@mean(EXPER) +
eq_logit.c(4)*@mean(EXPER)^2 + eq_logit.c(5)*@mean(AGE) +
eq_logit.c(6)*@mean(KIDSLT6) + eq_logit.c(7)*@mean(KIDSGE6) + eq_logit.c(8))
```

  - Recalculate, increasing the education by 1:

```
scalar ave_p_educ = @logit(eq_logit.c(1)*@mean(NWIFEINC) +
eq_logit.c(2)*(@mean(EDUC)+1) + eq_logit.c(3)*@mean(EXPER) +
eq_logit.c(4)*@mean(EXPER)^2 + eq_logit.c(5)*@mean(AGE) +
eq_logit.c(6)*@mean(KIDSLT6) + eq_logit.c(7)*@mean(KIDSGE6) + eq_logit.c(8))
```

  - Take the difference: `scalar mem_educ = ave_p_educ - ave_p_mean`

MEM_EDUC = 0.681446 – 0.631640 = 0.049807

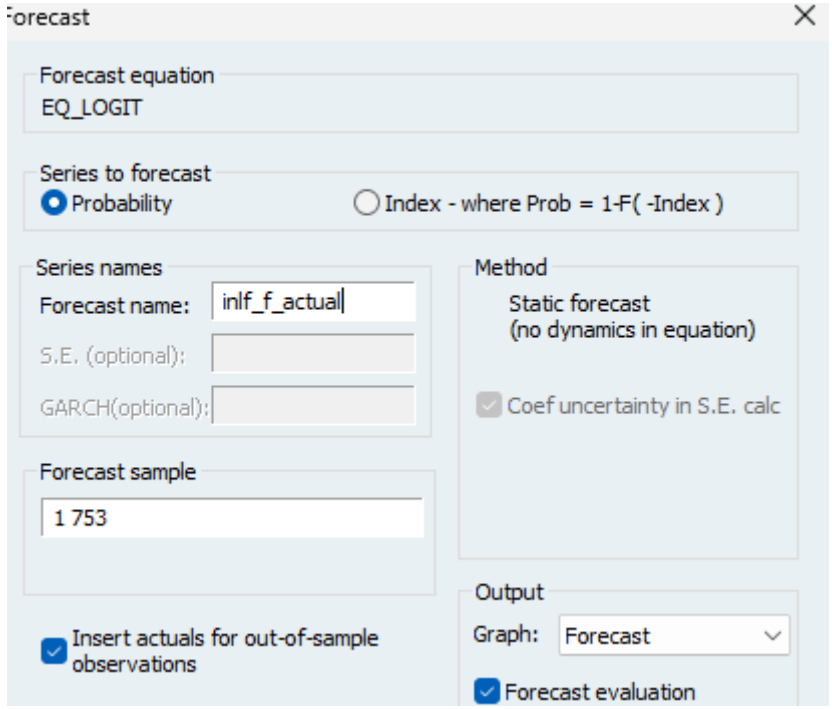
Use @cnorm for probit model instead of @logit.

Interpretation of the estimates in logit and probit. Average partial effect

- Calculate the average partial effect (APE).
  - **Predict at actual data.** For every observation, compute the fitted probability ($\hat{p}_i{}^0$) using its own values of all regressors
  - **Predict with one variable shifted.** Repeat the prediction ($\hat{p}_i{}^1$), adding one unit to the explanatory variable under examination only – all other regressors stay at their actual values.
  - **Average the individual differences.** APE $= \frac{1}{n} \sum_{i=1}^{n} (\hat{p}_i{}^1 - \hat{p}_i{}^0)$

Interpretation of the estimates in logit and probit. Average partial effect in EViews

1) Forecast for actual values



2) Rewrite the education series adding one to each observation (don't forget to save the actual values)

series educ = educ + 1

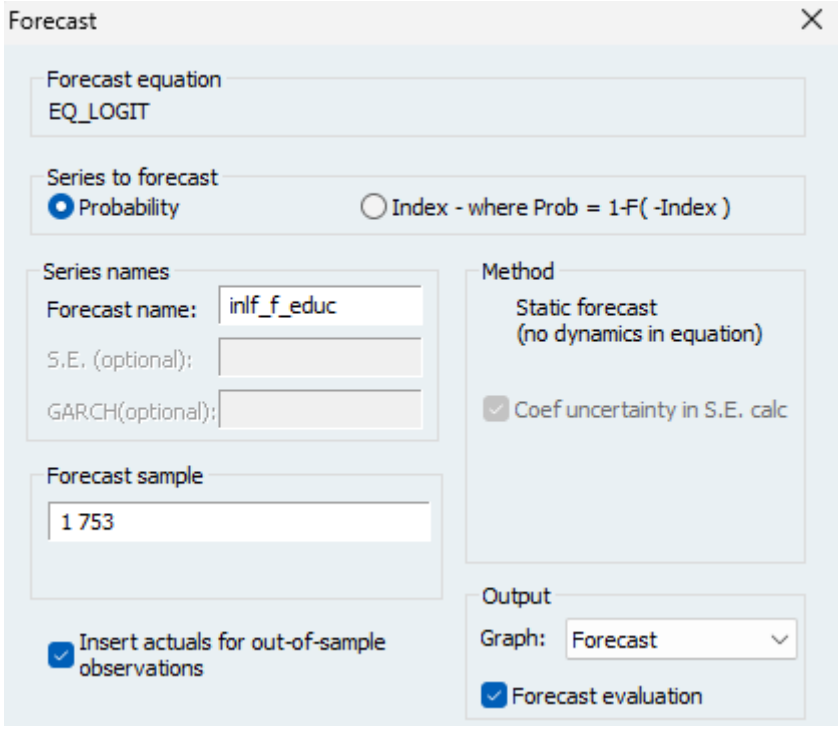
4) Calculate the mean of the differences in the predicted probabilities

scalar ape_educ = @mean(inlf_f_educ - inlf_f_actual)

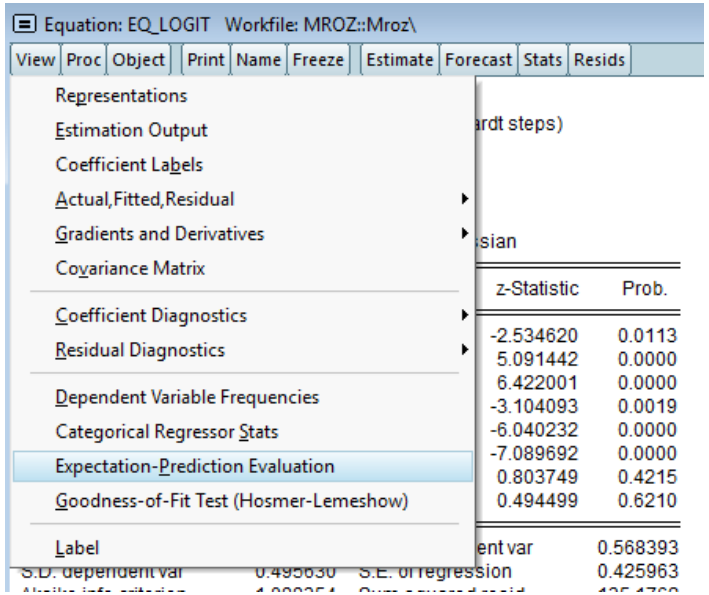
APE_EDUC = 0.038983

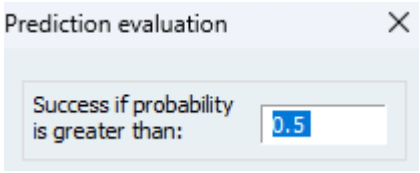
5) Restore the data (original education series)

3) Forecast for the adjusted education and the actual values of other regressors



Percent correctly predicted





Probit

Expectation-Prediction Evaluation for Binary Specification
Equation: EQ PROBIT
Date: 07/27/26 Time: 13:39
Success cutoff: C = 0.5

	Estimated Equation			Constant Probability		
	Dep=0	Dep=1	Total	Dep=0	Dep=1	Total
P(Dep=1)<=C	205	80	285	0	0	0
P(Dep=1)>C	120	348	468	325	428	753
Total	325	428	753	325	428	753
Correct	205	348	553	0	428	428
% Correct	63.08	81.31	73.44	0.00	100.00	56.84
% Incorrect	36.92	18.69	26.56	100.00	0.00	43.16

Logit

Expectation-Prediction Evaluation for Binary Specification
Equation: EQ_LOGIT
Date: 07/27/26 Time: 13:39
Success cutoff: C = 0.5

	Estimated Equation			Constant Probability		
	Dep=0	Dep=1	Total	Dep=0	Dep=1	Total
P(Dep=1)<=C	207	81	288	0	0	0
P(Dep=1)>C	118	347	465	325	428	753
Total	325	428	753	325	428	753
Correct	207	347	554	0	428	428
% Correct	63.69	81.07	73.57	0.00	100.00	56.84
% Incorrect	36.31	18.93	26.43	100.00	0.00	43.16

Logit and probit results are practically identical – the choice between them rarely matters in practice.

Summary

Logit and probit models are used for binary response variables.

These models have some advantages over the linear probability model: fitted probabilities are between zero and one, and the partial effects diminish.

The primary cost to logit and probit is that they are harder to interpret. To recover probabilities, compute either the partial effect at the mean (one hypothetical average person) or the average partial effect (averaged over the actual sample). The APE is normally the policy-relevant figure, and it lands close to the LPM coefficient